

Quiz 6: Computer Networks

ID:

Total Marks: 20

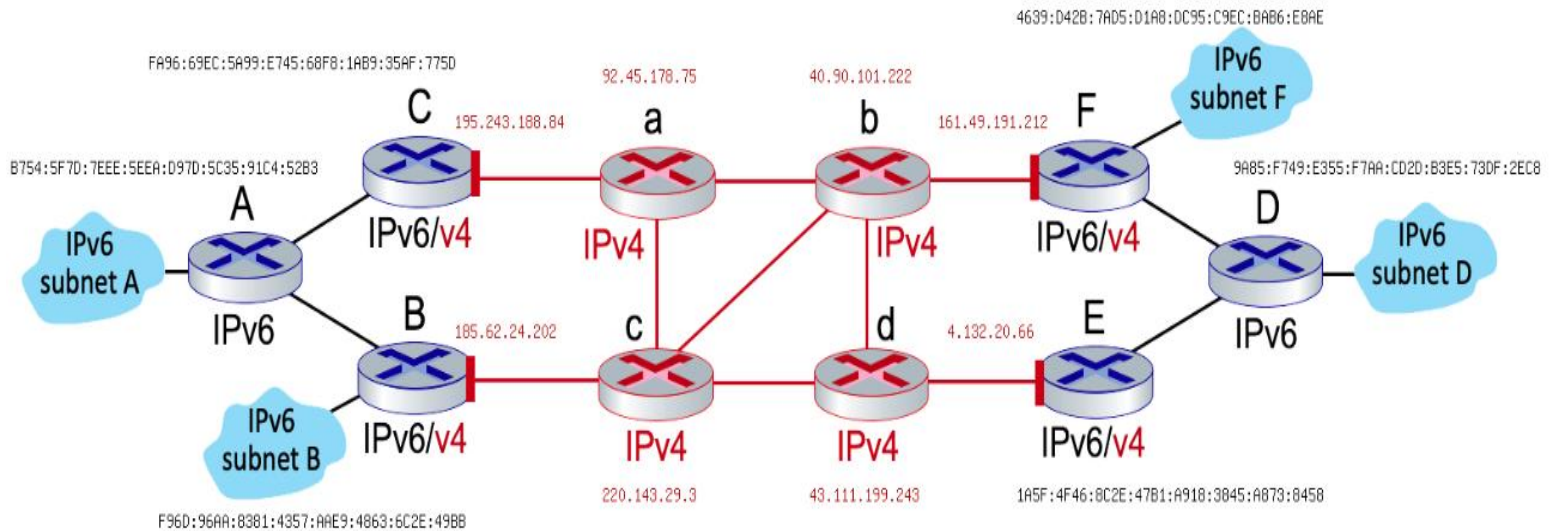
Name:

Group A

Section:

Total Marks: [20 Marks]

Scenario



Suppose that a host of subnet F wants to send an IPv6 datagram to a host on subnet A. Assume that the forwarding between these two hosts goes along the path: F --> b --> d --> c --> B --> A

1. What is the destination address of this b to d datagram?

The destination IP address is 185.62.24.202

2. What router is the 'tunnel entrance'? Give the router's letter

The tunnel entrance is router F

3. What router is the 'tunnel exit'? Give the router's letter

The tunnel exit is router B

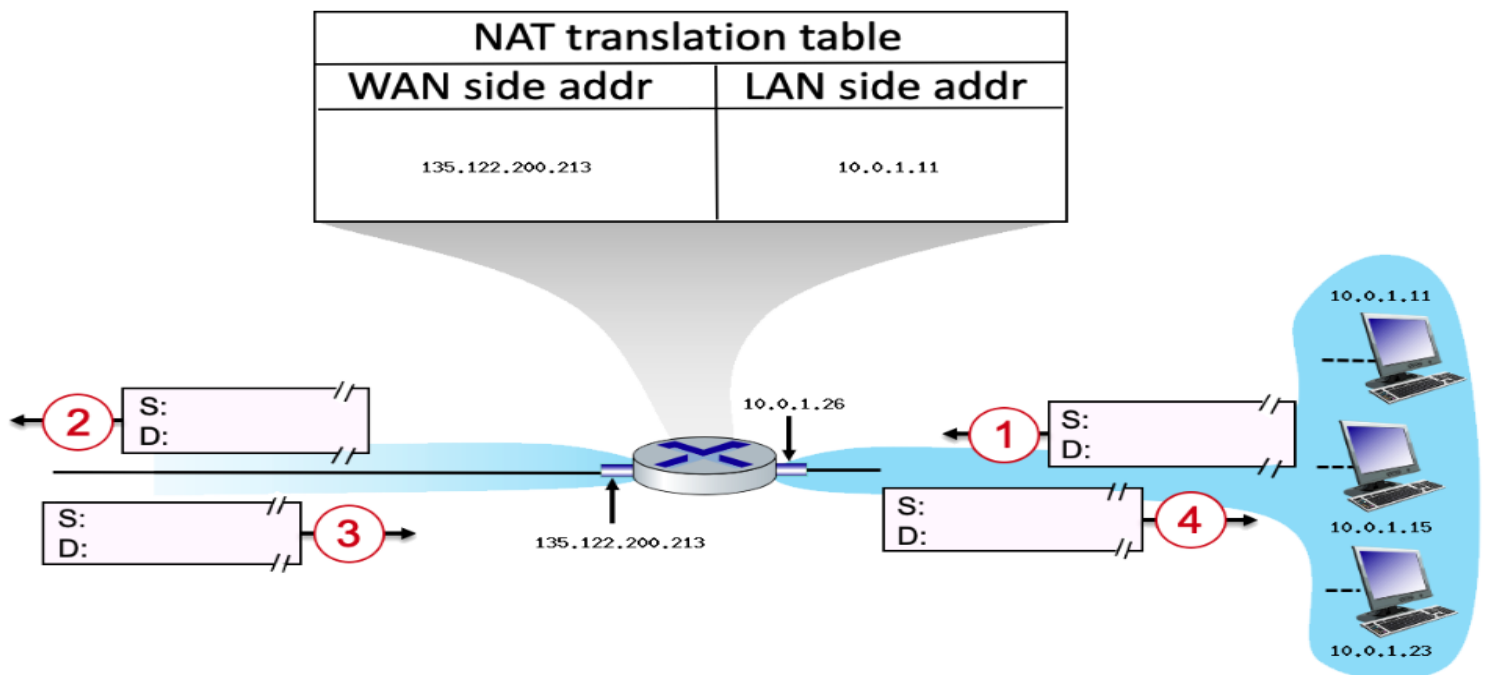
4. Is this c to B datagram encapsulating another datagram? Yes or No

yes

5. What is the destination address of this B to A datagram?

The destination IP address is B754:5F7D:7EEE:5EEA:D97D:5C35:91C4:52B3

Scenerio



Suppose that the host with IP address 10.0.1.11 sends an IP datagram destined to host 128.119.165.181. The source port is 3426, and the destination port is 80.

1. At step 1, what is the destination IP address?
The destination address will be the remote machine's IP, which is 128.119.165.181
2. consider the datagram at step 2, after it has been transmitted by the router. What is the source IP address for this datagram?
The source address will be the router's public IP, which is 135.122.200.213
3. Will the source port have changed? Yes or No.
Yes, the NAT will change the source port.
4. consider the datagram at step 4, after it has been transmitted by the router but before it has been received by the host. What is the source IP address for this datagram?
The source address will be the remote machine's IP, which is 128.119.165.181
5. At step 4, what is the destination IP address for this datagram
The destination address will be the local host's IP, which is 10.0.1.11